

① Find mean, median & mode of the data

85, 75, 30, 86, 85,

Step 1:- Arrange in ascending order: 30, 30, 35, 45, 75, 85, 85, 85, 86

$$\text{Mean} = \frac{\sum x_i}{N} = \frac{30+30+35+45+75+85+85+85+86}{9}$$

$$\text{Mean} = 556/9 = 61.78$$

Median $\rightarrow$ (middle value when arrange in ascending order)

$$n=9, \text{ median} = \frac{(9+1)}{2} = \frac{10}{2} = 5^{\text{th}} \text{ value}$$

5th value in ordered data = 75 = Median

Mode (most frequent data): 85 occurs 3 times.

$$\text{Mode} = 85.$$

② Find 5 number summary on the following data:-

Data: 34, 18, 100, 27, 54, 52, 93, 59, 61, 87, 68, 85, 78, 82, 91.

Step 1:- Arrange in ascending order.

18, 27, 34, 52, 54, 59, 61, 68, 78, 82, 85, 87, 91, 93, 100.

Five number summary:-

① minimum (lowest value) = 18

② maximum (highest value) = 100

③ Median (Q_2): Since there are 15 value ($n=15$, odd)

$$\text{Median} = \frac{(n+1)^{\text{th}}}{2} = \frac{16}{2} = 8^{\text{th}} \text{ value} = 68$$

④ First quartile (Q_1): $\left(\frac{(n+1)}{4}\right)^{\text{th}}$ observation

Q_1 = median of lower half (first 7 values):-

18, 27, 34, 52, 54, 59, 61

$$\text{Median} = 4^{\text{th}} \text{ value} = 52 = Q_1$$

⑤ Third quartile (Q_3): $\left(\frac{3(n+1)}{4}\right)^{\text{th}}$ observation

Q_3 = median of upper half (last 7 values)

78, 82, 85, 87, 91, 93, 100

$$\text{Median} = 4^{\text{th}} \text{ value} = 87 = Q_3.$$

Q3: - Find the outlier using whisker plot (2)
 60, 82, 82, 88, 84, 92, 90, 90, 97, 93
 range of data values that are not outliers

① Arrange in ascending order
 60, 82, 82, 84, 88, 90, 90, 92, 93, 97

② Find quartiles $n=10$ (even) $\Rightarrow \frac{\text{value at } (n/2)^{\text{th}} + \text{value at } (n/2+1)^{\text{th}}}{2}$
 Median (Q_2) = average of 5th & 6th value
 $= \frac{(88+90)}{2} = 89$

Q1 (first quartile):
 Lower half = 60, 82, 82, 84, 88
 Median = 3rd value = 82

$$10/2 \Rightarrow 5 \quad \frac{10+1}{4} = \frac{11}{4} \Rightarrow 2.75$$

Q3 (third quartile):
 Upper half = 90, 90, 92, 93, 97
 Median = 3rd value = 92

$$\frac{3(10+1)}{4} = \frac{33}{4} = 8.25 \Rightarrow 8.1$$

Step 3: - Calculate IQR (Interquartile Range)
 $IQR = Q_3 - Q_1 = 92 - 82 = 10$

Step 4: - Find whiskers (fences)
 Lower fence = $Q_1 - 1.5 \times IQR$
 $= 82 - 1.5(10) = 67$

Upper fence = $Q_3 + 1.5 \times IQR = 92 + 15 = 107$

Steps: - Identify outliers

Any value < 67 or > 107 is outlier

Data = 60, 82, 82, 84, 88, 90, 90, 92, 93, 97

Only $60 < 67 \rightarrow$ outlier = 60

$\rightarrow$ An outlier is a data point that is significantly different from other values in dataset.
 It is much smaller or larger compared to most of data.

:- find the intercept & slope in linear regression based on following data:-

X	1	3	4	5	8
Y	5	3	2	0	0

or.
find regression of Y on X.

Eqⁿ of regression line:-
 $Y = a + bX$

$a \rightarrow$ intercept
 $b \rightarrow$ slope

① Compute mean

$$\bar{X} = \frac{1+3+4+5+8}{5} = \frac{21}{5} = 4.2$$

$$\bar{Y} = \frac{5+3+2+0+0}{5} = 10/5 = 2$$

$$b = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sum (X - \bar{X})^2}$$

X	Y	$(X - \bar{X})$	$(Y - \bar{Y})$	$(X - \bar{X})(Y - \bar{Y})$	$(X - \bar{X})^2$
1	5	-3.2	3	-9.6	10.24
3	3	-1.2	1	-1.2	1.44
4	2	-0.2	0	0	0.04
5	0	0.8	-2	-1.6	0.64
8	0	3.8	-2	-7.6	14.44
				<u>-20.0</u>	<u>26.8</u>

$$b = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sum (X - \bar{X})^2} = \frac{-20.0}{26.8} = -0.746$$

$$a = \bar{Y} - b\bar{X} = 2 - (-0.746 \times 4.2)$$

$$a = 2 + 3.133 = 5.133$$

Final Regression eqⁿ $\Rightarrow Y = 5.133 - 0.746X$

Intercept (a) = 5.133, slope (b) = -0.746

⇒ Normalised the Data = [15, ...]
Sol Apply standard

Ques 4 Significance of Correlation. Calculate coefficient of correlation between height (in inches) of father and height of son (in inches) from data: (3)

height of father	66	67	68	69	70	71	72
height of son	68	69	67	69	72	70	74

→ Correlation measures strength & direction of linear relationship between two variables.

Significance: - helps in prediction

- detects degree of association between variables
- important for feature selection
- strength of relationship
- measure correlation between variables

Pearson Coefficient of Correlation (r): -

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \cdot \sum (y_i - \bar{y})^2}}$$

⇒ Normalized the data = [15, 25, 30, 40, 50]
 So Apply standard normalization:-

$$Z = \frac{x - \mu}{\sigma}$$

x = value

μ = mean of data

σ = standard deviation of data

① Find mean ⇒

$$\mu = \frac{15 + 25 + 30 + 40 + 50}{5} = \frac{160}{5} = 32$$

② Find standard deviation (σ)

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n}}$$

$$(15 - 32)^2 = (-17)^2 = 289, (25 - 32)^2 = (-7)^2 = 49$$

$$(30 - 32)^2 = (-2)^2 = 4, (40 - 32)^2 = 8^2 = 64, (50 - 32)^2 = 324$$

$$\text{sum} = 289 + 49 + 4 + 64 + 324 = 730$$

$$\sigma = \sqrt{\frac{730}{5}} = \sqrt{146} \approx 12.083$$

③ Apply standard normalization:- $Z = \frac{x - \mu}{\sigma}$

$$Z_{15} = \frac{15 - 32}{12.083} = -1.408,$$

$$Z_{25} = \frac{25 - 32}{12.083} \approx -0.579$$

$$Z_{30} = \frac{30 - 32}{12.083} = \frac{-2}{12.083} = -0.165,$$

$$Z_{40} = 0.662$$

$$Z_{50} = \frac{50 - 32}{12.083} = 1.491$$

$$\text{Normalized data} = [-1.408, -0.579, -0.165, 0.662, 1.491]$$

Q1 min max normalization:-

$$x_{\min} = 10, x_{\max} = 50$$

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

$$x'_{10} = \frac{10 - 10}{50 - 10} = \frac{0}{40} = 0,$$

$$x'_{20} = \frac{20 - 10}{50 - 10} = \frac{10}{40} = 0.25,$$

$$x'_{30} = 0.5$$

$$x'_{40} = 0.75$$

$$\text{Normalized data (min-max)} = [0, 0.25, 0.5, 0.75, 1]$$

Q1 Compute absolute error, mean squared error & root mean square error based on data:-

Observed value (actual)	10	15	20	25	30
Predicted val.	12	18	22	24	28

① Absolute error (AE)

$$AE_i = |\text{Observed}_i - \text{Predicted}_i|$$

$$AE_1 = |10 - 12| = 2, \quad AE_2 = |15 - 18| = 3$$

$$AE_3 = |20 - 22| = 2, \quad AE_4 = |25 - 24| = 1, \quad AE_5 = |30 - 28| = 2$$

$$\text{Absolute errors} = [2, 3, 2, 1, 2]$$

② Compute mean square error (MSE) :-

$$MSE = \frac{\sum (\text{Observed}_i - \text{Predicted}_i)^2}{n}$$

$$(10 - 12)^2 = (-2)^2 = 4, \quad (15 - 18)^2 = (-3)^2 = 9$$

$$(20 - 22)^2 = (-2)^2 = 4, \quad (25 - 24)^2 = 1^2 = 1$$

$$(30 - 28)^2 = (2)^2 = 4$$

$$\text{Sum of squares} = 4 + 9 + 4 + 1 + 4 = 22$$

$$MSE = 22/5 = 4.4$$

③ Root mean square error (RMSE)

$$RMSE = \sqrt{MSE} = \sqrt{4.4} \approx 2.097$$